Handheld: pairing and warehouse work

This is an informational translation. The matching Russian revision is authoritative. Step-by-step warehouse instruction for Markiro Autonom: pairing a handheld terminal with the host by a code from the office, operator sign-in, the label printer, building a pallet from boxes, printing labels, disassembling packs, reprinting labels and working without a link to the host.

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1. Purpose

This instruction covers the warehouse «Палеты» (Pallets) mode on the handheld terminal (ТСД) in Markiro Autonom. The operator scans finished boxes' SSCC labels; the handheld closes the pallet and prints its label on the assigned printer. Production-shift work on the handheld is outside this instruction's scope. Line-station work is covered in MKR-INS-01–03.

This guide covers Handheld 1.11.0 with Markiro Autonom host 1.7.0. The handheld connects to the host over the local network. Warehouse work does not require internet access. When the host is offline, pallet building can continue with the downloaded registry and remaining SSCCs.

Note: The Markiro Autonom interface is in Russian only, so the screenshots are Russian and screen labels are quoted in Russian with an English translation in brackets. The screenshots use demonstration data; addresses, codes, the host fingerprint and SSCC numbers on your site will differ.

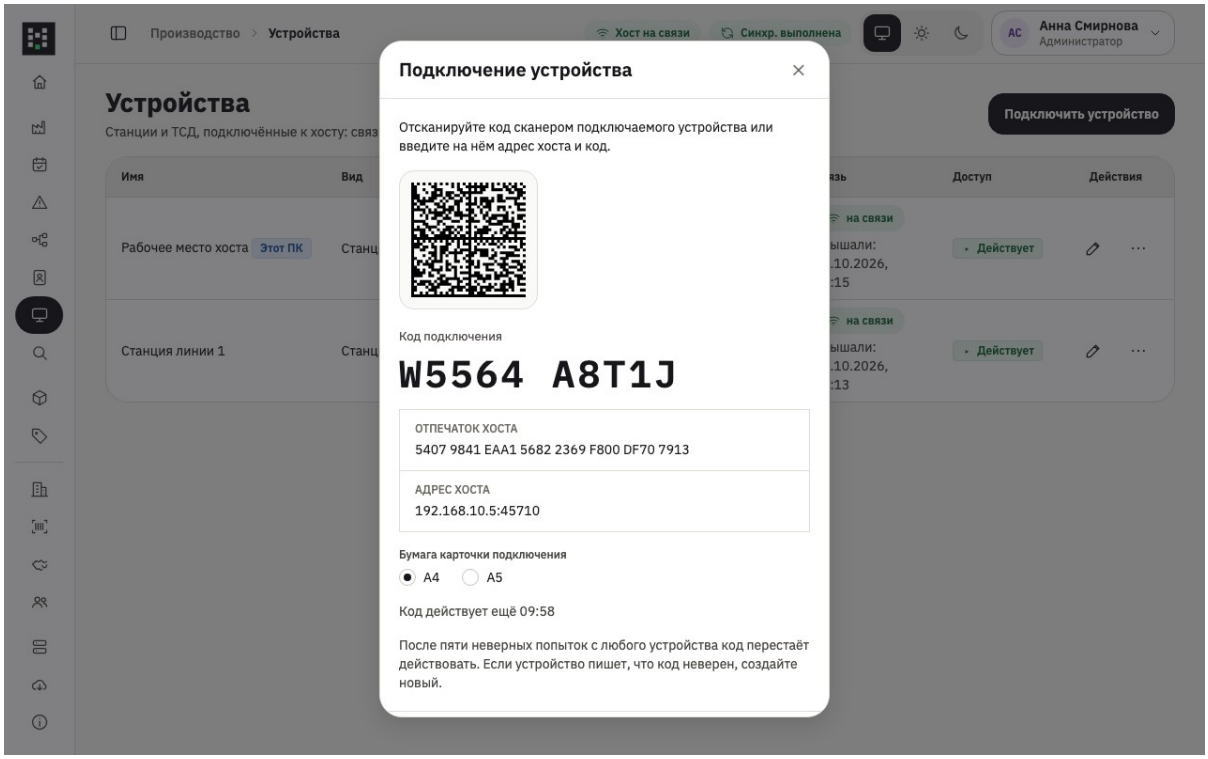
2. Preparation

- The host is installed and running, and the handheld is on the same Wi-Fi network as the host computer.
- The host computer has a permanent network address: the handheld remembers the address when it is paired. Reserving an address is described in the host installation instruction.
- On the host computer the Wi-Fi network is marked as private, and client isolation is off on the access point; otherwise the handheld cannot see the host although both are on the network.
- The clocks of the handheld and the host computer agree within two minutes.
- You have access to the Markiro Autonom office as an administrator: the pairing code is created in «Устройства» (Devices).
- The label printer is reachable over Wi-Fi or Bluetooth, and you know its address, language (ZPL or TSPL) and resolution.

3. Pairing the handheld with the host

Step 1. 3.1. Create a code in the office

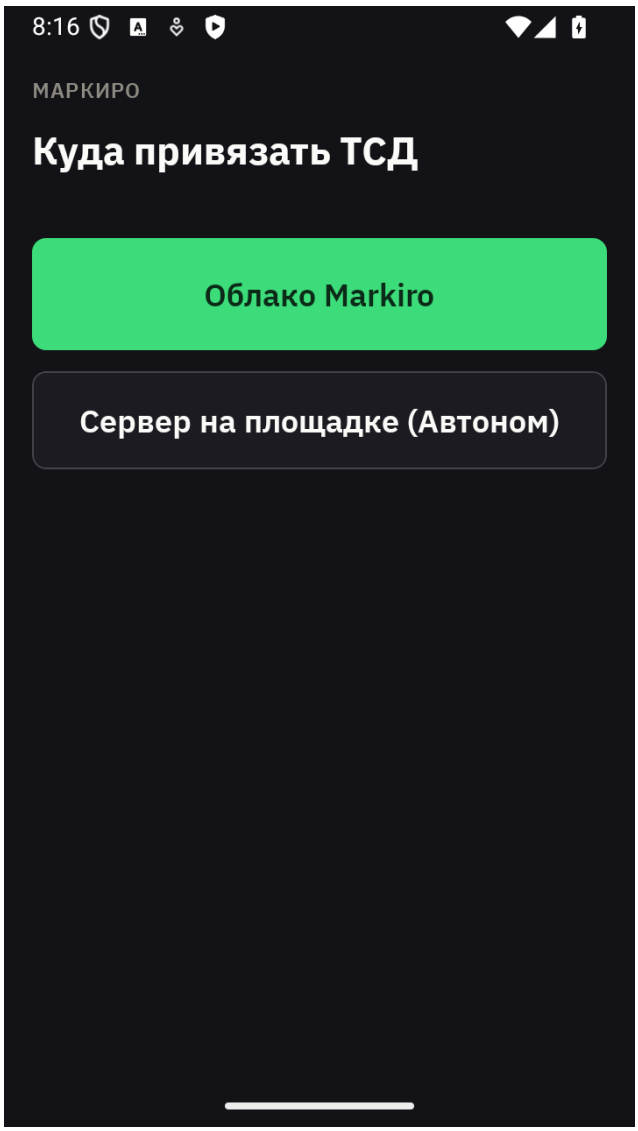
In the office open «Устройства» (Devices) and press «Подключить устройство» (Connect a device). The office shows the pairing barcode, the 10-character «Код подключения» (Pairing code), and the host's fingerprint and address. The code is single-use; the time left is shown under it, and «Новый код» (New code) issues another one.



Office: the pairing code for a new device

Step 2. 3.2. Choose the on-site server

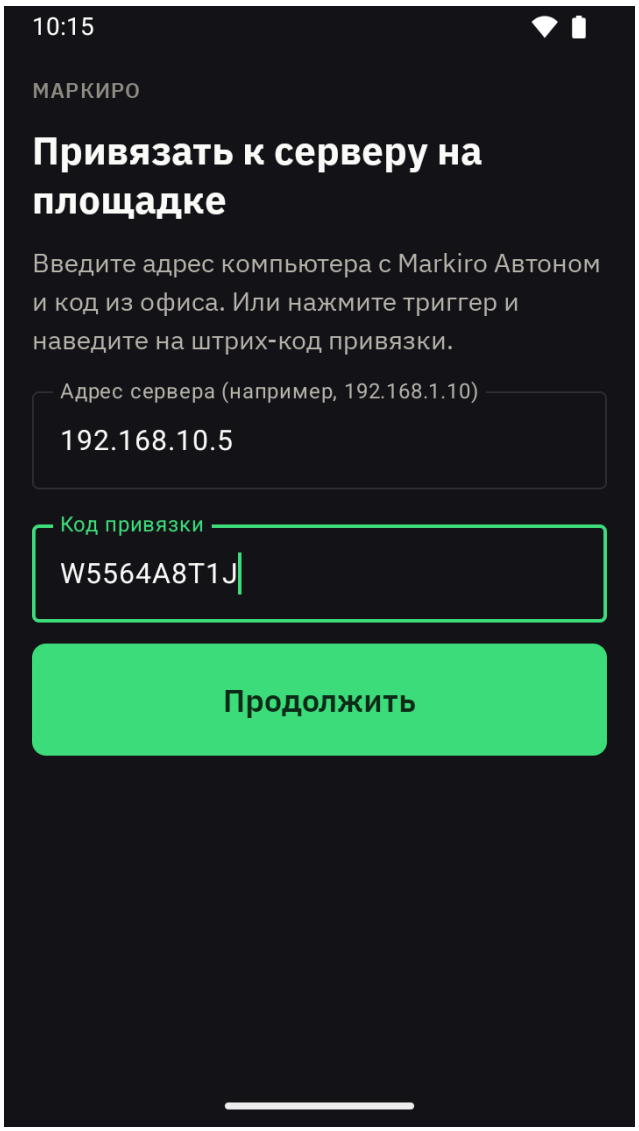
Start the app on the handheld. On «Куда привязать ТСД» (Where to pair the handheld) press «Сервер на площадке (Автоном)» (On-site server (Autonom)).



Handheld: choosing where to pair the device

Step 3. 3.3. Scan the barcode or type the address and code

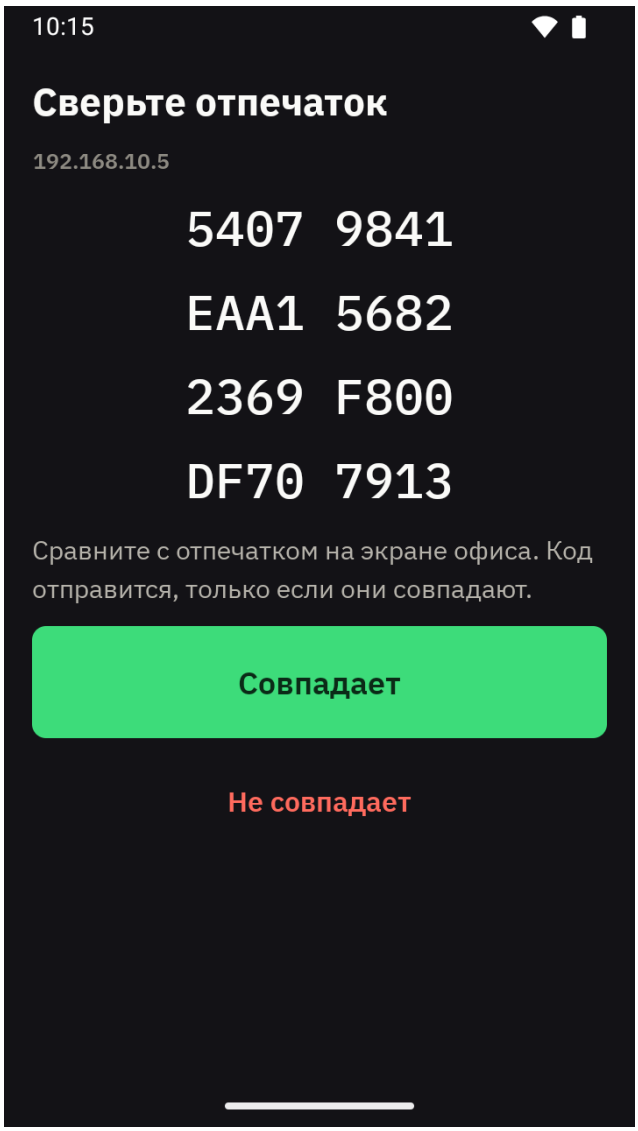
The simplest way is to press the trigger and aim the handheld at the pairing barcode on the office screen. Otherwise type «Адрес сервера» (Server address), the host address from the office without the port, for example 192.168.1.10, and «Код привязки» (Pairing code), then press «Продолжить» (Continue).



Handheld: the host address and pairing code typed by hand

Step 4. 3.4. Compare the fingerprint

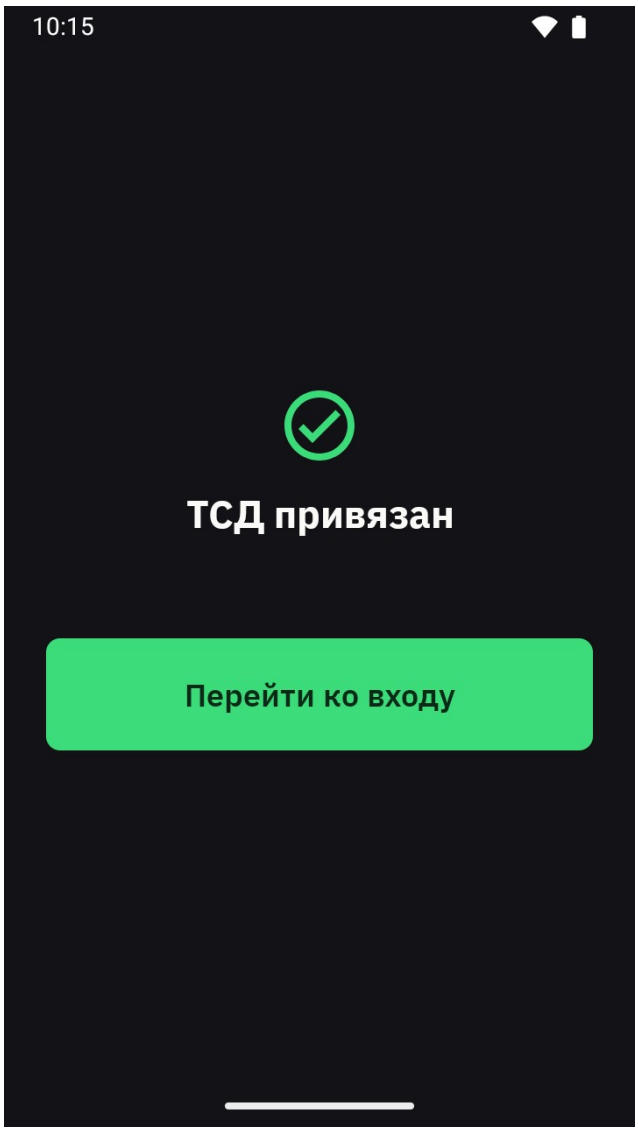
After manual entry the handheld shows «Сверьте отпечаток» (Compare the fingerprint) with the fingerprint of the host at that address. Compare it with the fingerprint on the office screen: the code is sent to the host only if they match. Press «Совпадает» (Matches); if the fingerprint differs, press «Не совпадает» (Does not match) and check the address. When the barcode is scanned, the handheld checks the fingerprint itself.



Handheld: comparing the host fingerprint

Step 5. 3.5. Wait for pairing to finish

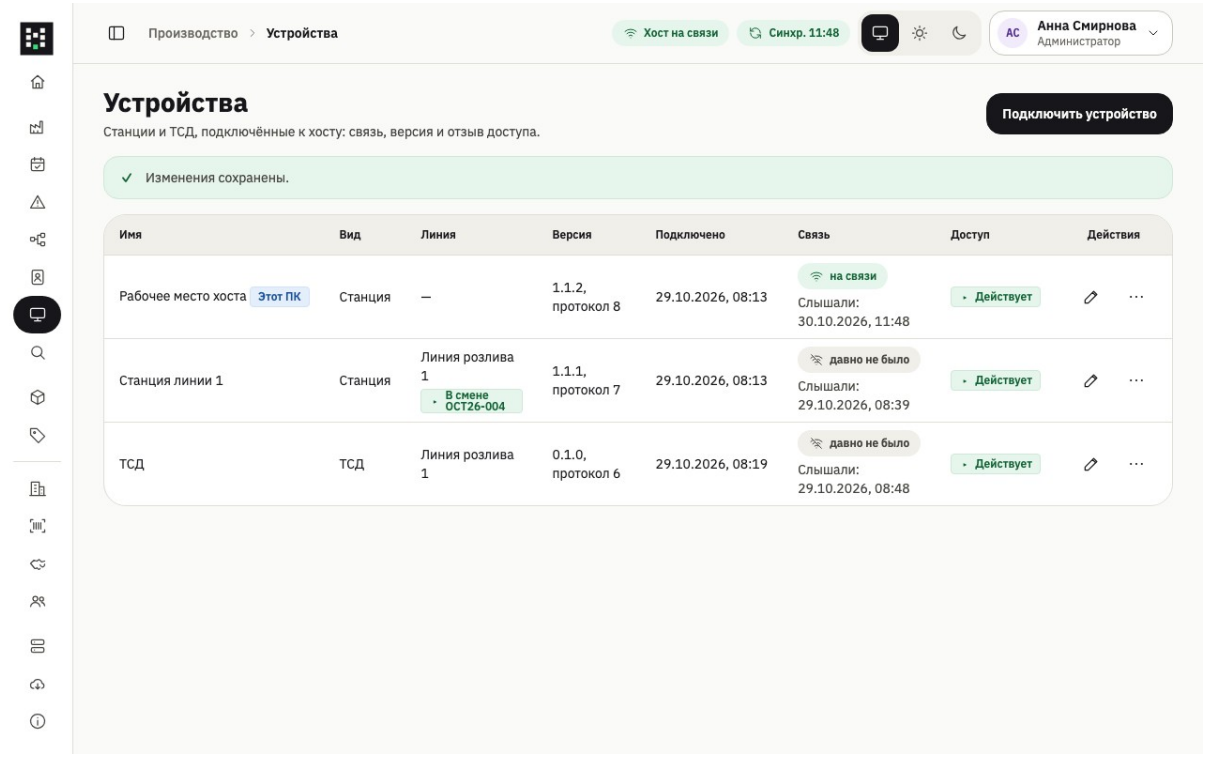
The handheld loads the reference data from the host and shows «ТСД привязан» (Handheld paired). Press «Перейти ко входу» (Go to sign-in).



The handheld is paired with the host

Step 6. 3.6. Check the device in the office

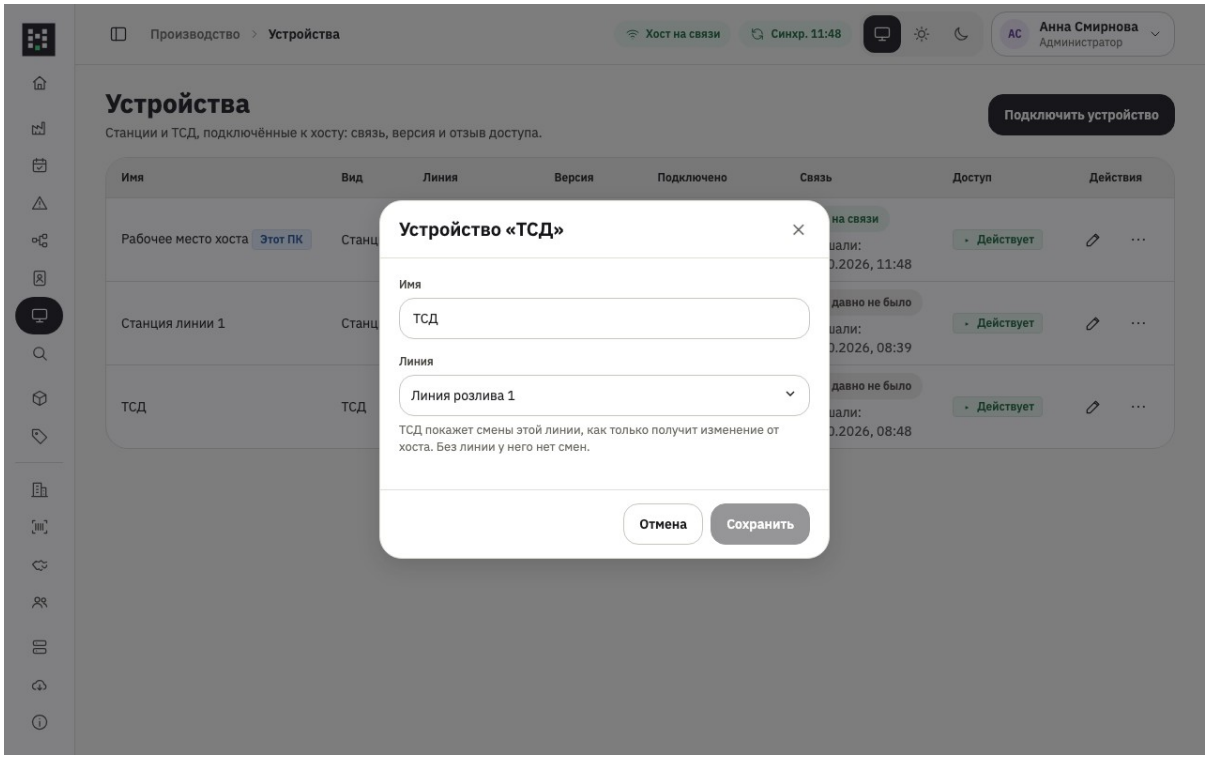
The office shows «Устройство «ТСД» подключено.» (Device “ТСД” connected.), and the «Устройства» (Devices) list gets a row of kind «ТСД» (Handheld). The device name can be changed with the pencil button in its row.



Office: the handheld in the device list

Step 7. 3.7. Assign a line if needed

If the handheld will work in a production shift, the administrator presses the pencil in its device row, chooses «Линия» (Line), and saves the change. The handheld receives the line when connected to the host; the «Смена» (Shift) tile lists that line's planned and active shifts. The warehouse «Палеты» (Pallets) mode needs no line. If the «Линия» field is absent, update the office PC application to a version that supports handheld line assignment.



Office: assigning the handheld's line

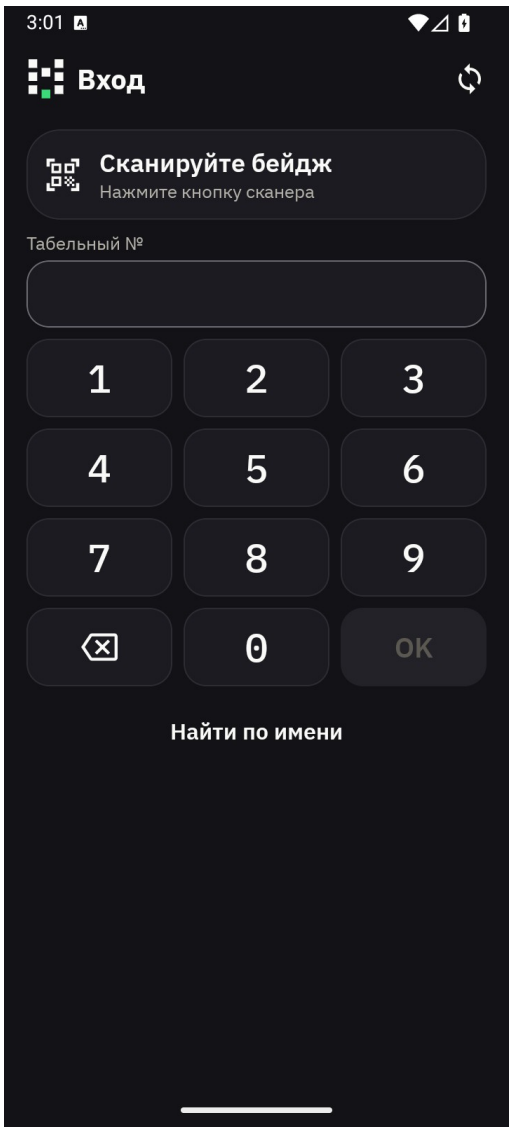
Message on the handheld	What to do
«Срок кода истёк. Выпустите новый код в офисе.» (The code has expired. Issue a new code in the office.)	In the office press «Новый код» (New code) and pair again.
«Код не подошёл. Выпустите новый код в офисе.» (The code did not match. Issue a new code in the office.)	Check the code against the office screen or issue a new one.
«Сервер не отвечает…» (The server does not respond…)	Check Wi-Fi on the handheld, the host address, the network profile and client isolation on the access point.
«В лицензии нет свободных мест для ТСД.» (The licence has no free handheld seats.)	The licence sets the number of handhelds; contact the administrator.
«Работа с ТСД не включена в офисе.» (Handheld work is not enabled in the office.)	Contact the administrator.

If pairing fails

4. Operator sign-in

Step 1. Sign in with a badge or personnel number

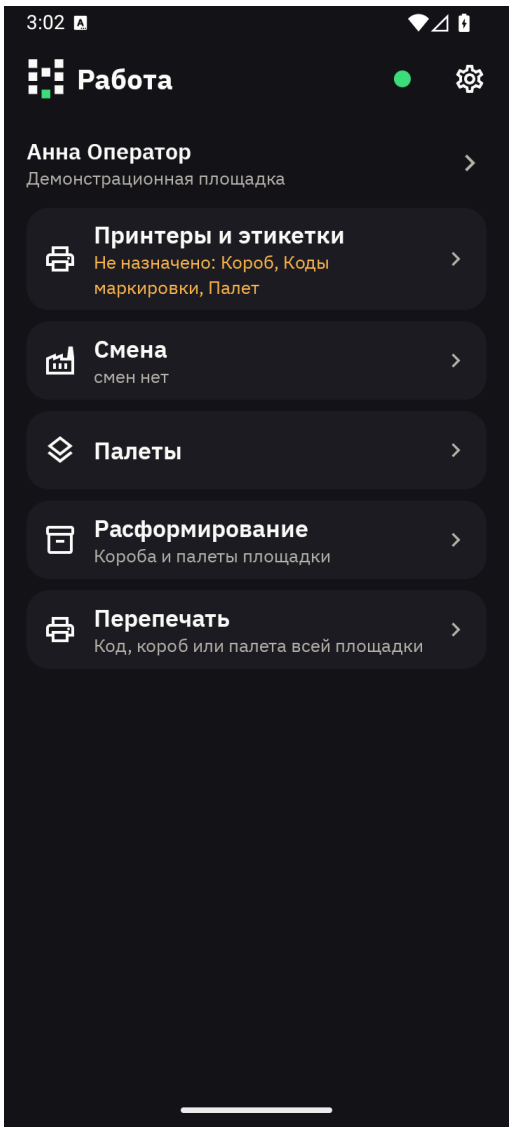
On the sign-in screen scan your badge, or type «Табельный №» (Personnel No.) on the on-screen keypad and then your PIN. If you do not remember the number, press «Найти по имени» (Find by name). The handheld stores the operator list for offline sign-in. When connected, tap «Обновить список» (Refresh list) in the header to reload it.



Handheld: operator sign-in

Step 2. The main screen

After sign-in, «Работа» (Work) opens. The menu includes «Смена» (Shift), «Палеты» (Pallets), «Расформирование» (Disassembly) and «Перепечать» (Reprint). The status indicator opens host connection and queue details, and the gear opens settings. To end the session, tap the operator name and choose «Сменить оператора» (Switch operator).

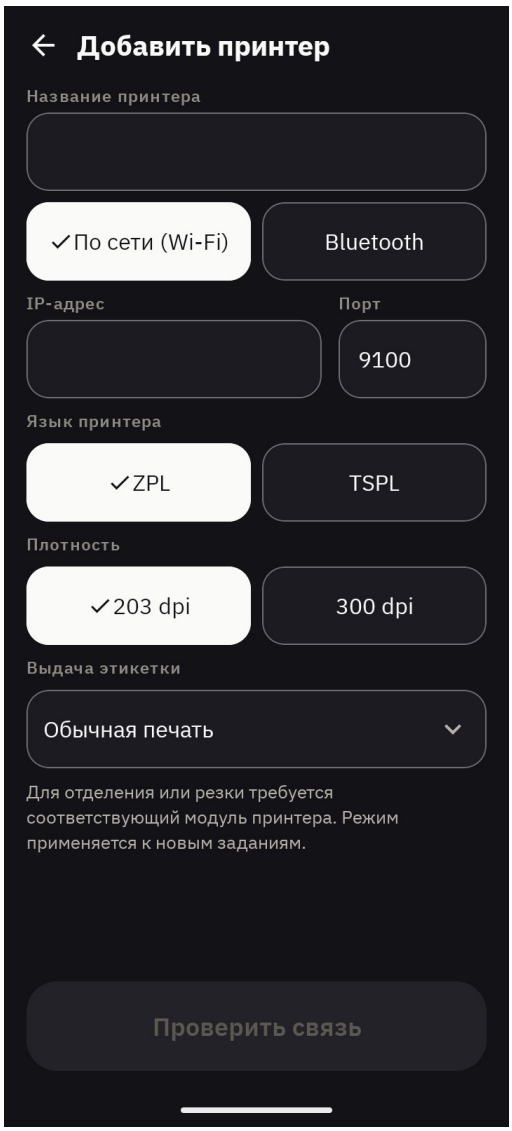


Handheld: the main screen without an assigned line

5. The label printer

Step 1. Add the printer

On «Работа» (Work), tap the gear. In Settings open «Принтеры и этикетки» (Printers and labels) → «Добавить по адресу» (Add by address). For a network printer choose «По сети (Wi-Fi)» (Network) and enter its name, IP address and port (usually 9100). Set its configured language (ZPL or TSPL) and density (203 or 300 dpi). Tap «Проверить связь» (Check connection) to save it.



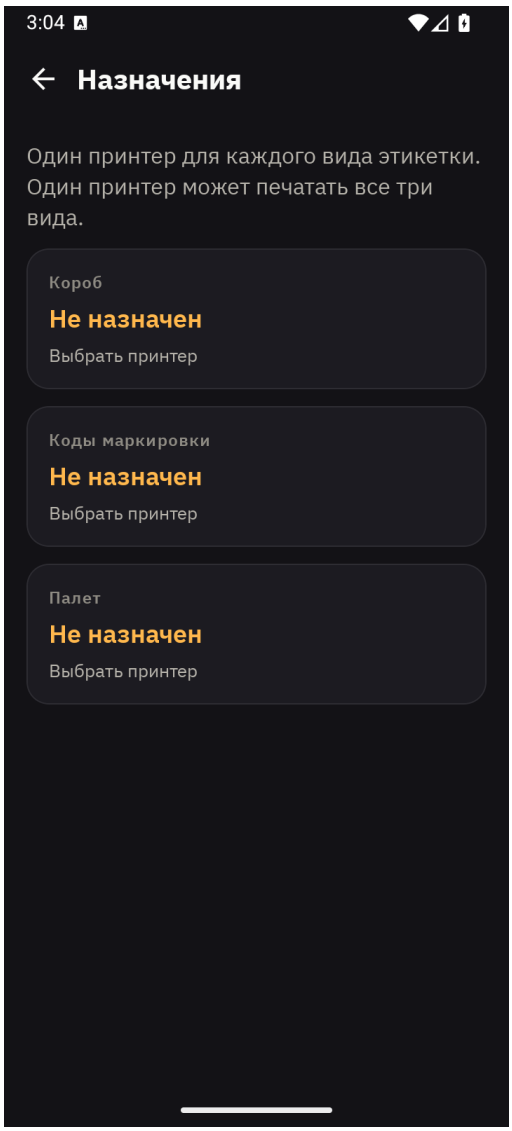
Handheld: adding a network printer

Step 2. Connect a Bluetooth printer

Pair the printer with the handheld in Android settings first. Then select «Bluetooth» in the add-printer form. On Android 12 or later, tap «Разрешить» (Allow) and grant Nearby devices access. If the system prompt no longer appears, tap «Открыть настройки приложения» (Open app settings) and grant access there. Choose the printer from the list and check the connection.

Step 3. Assign the printer

On «Принтеры» (Printers), open «Назначения» (Assignments). For warehouse pallet building, tap «Выбрать принтер» (Choose printer) in the «Палет» (Pallet) row. To reprint other label types, assign printers in the «Короб» (Box) and «Коды маркировки» (Marking codes) rows. One printer can serve all three.



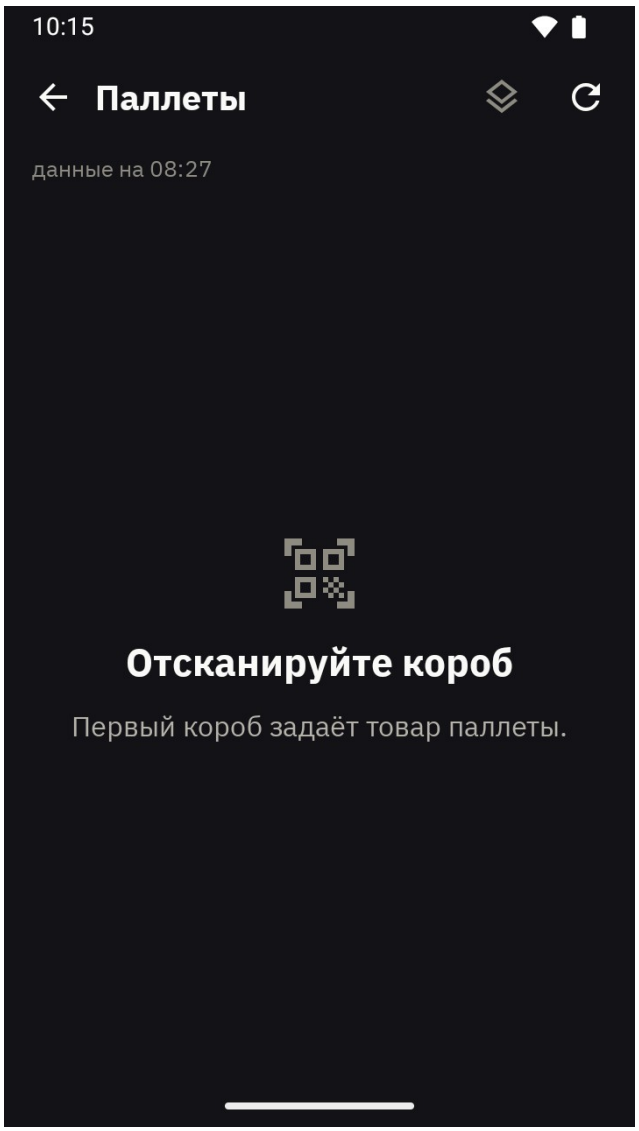
Handheld: printer assignment by label kind

In the printer list, tap «Тест» (Test) for the required printer. Inspect the printout and check that the barcode scans. To choose a different template, tap «Этикетки» (Labels) for the printer, choose the label type and check the preview size before assigning it. Queued jobs keep their existing templates. Set «Выдача этикетки» (Label output) to match the printer hardware. Peel-off and cutting require the corresponding modules.

6. Building a pallet

Step 1. Open pallet building

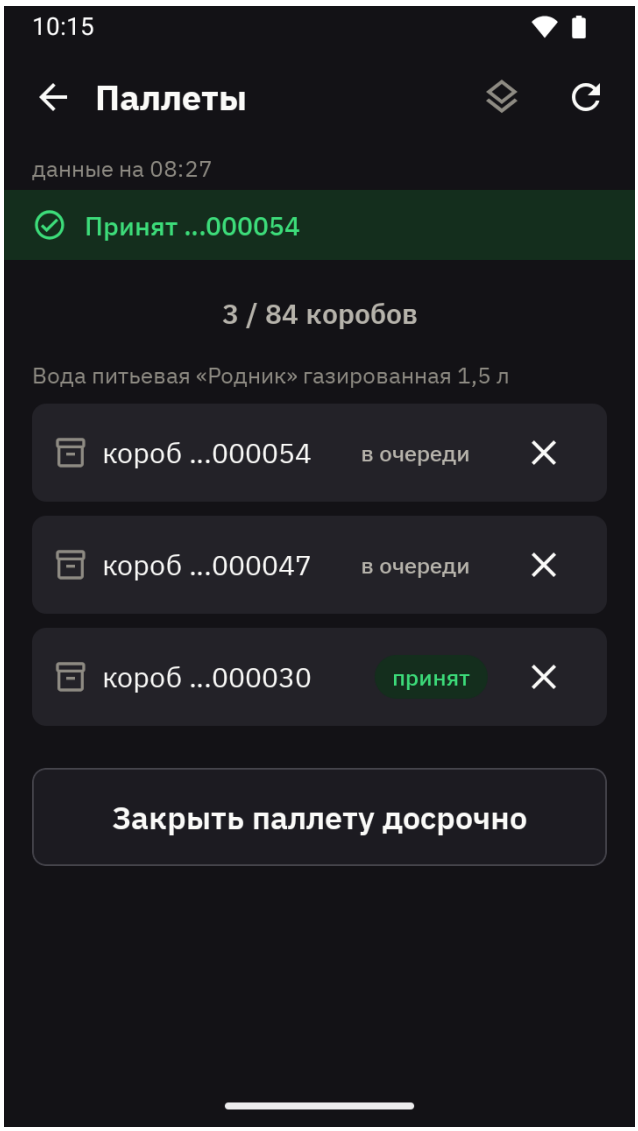
On the main screen press «Палеты» (Pallets). The handheld waits for the first box: «Отсканируйте короб» (Scan a box). The first box sets the pallet's product. The line «данные на …» (data as of …) shows when the handheld last loaded the box registry from the host; «Обновить реестр» (Refresh registry) at the top of the screen loads it again.



Handheld: a new pallet waits for the first box

Step 2. Scan the boxes' SSCC labels

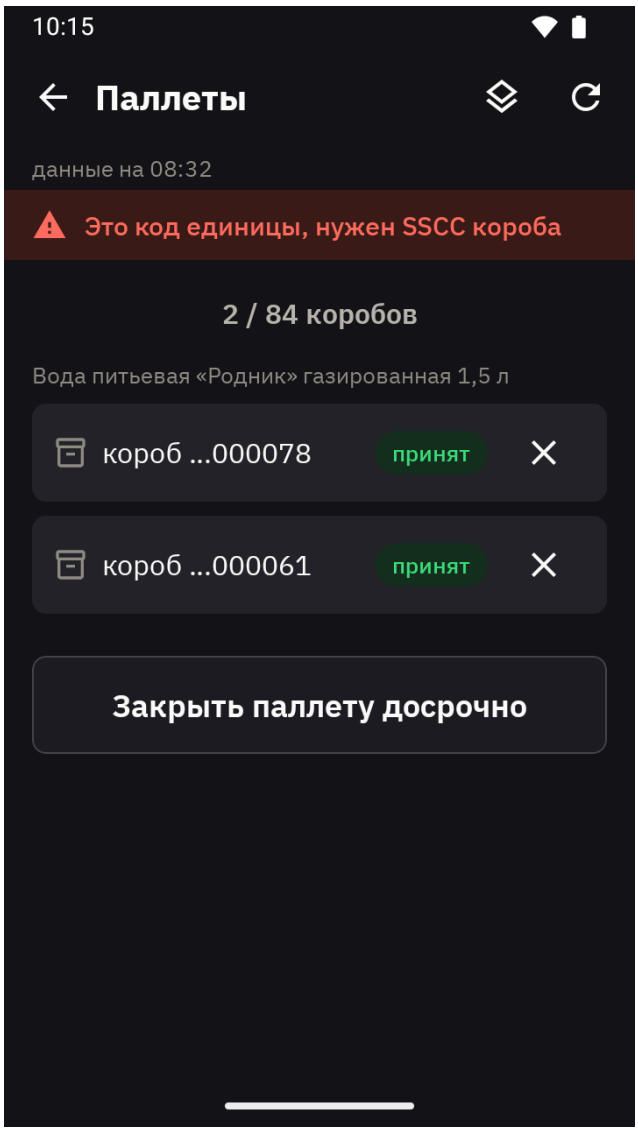
Scan the box labels one at a time. An accepted box appears in the list, the top line says «Принят» (Accepted) with the last digits of the SSCC, and above the list the handheld shows how many boxes are on the pallet out of the pallet capacity in the product card. Each box is first «в очереди» (queued) and then «принят» (accepted) by the host. A box put on by mistake is removed with the cross button in its row.



Handheld: three boxes on the pallet

Step 3. If a box is not accepted

When a scan does not fit, the handheld gives the reason in a red line at the top and does not put the box on the pallet. The most common messages are in the table below.



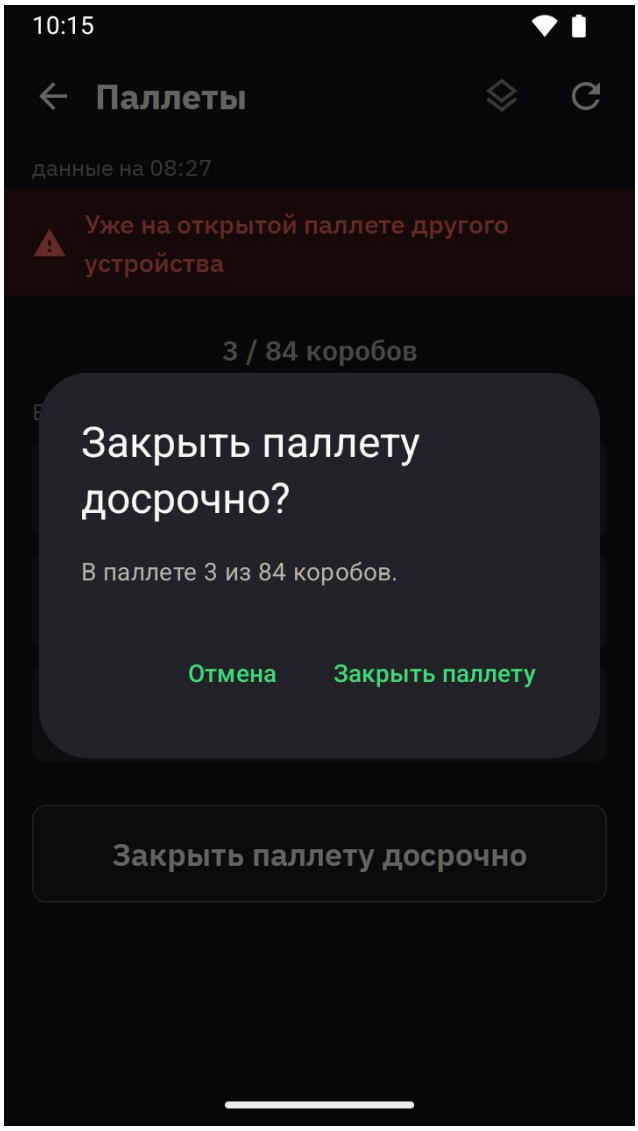
Handheld: a unit code scanned instead of a box SSCC

Message	What to do
«Уже на этом палете ...» (Already on this pallet ...)	Nothing: the box is already in the list.
«Уже на палете ...» (Already on pallet ...), «Уже на открытом палете другого устройства» (Already on another device's open pallet)	The box is recorded on another pallet. Set it aside and tell your supervisor.
«Уже на другом палете этого ТСД» (Already on another pallet of this handheld)	The box is on another pallet of this handheld. Set it aside and tell your supervisor.
«Другой товар: ...» (Another product: ...)	A pallet takes boxes of one product. Set the box aside.
«Короб неизвестен. Обновите реестр» (Unknown box. Refresh the registry)	Press «Обновить реестр» (Refresh registry) and scan the box again.
«Это код единицы, нужен SSCC короба» (This is a unit code; a box SSCC is needed)	Scan the box label, not the code on a unit.
«Это палет, не короб» (This is a pallet, not a box)	A pallet label was scanned. Scan the box label.

Messages when a box is scanned

Step 4. Close the pallet

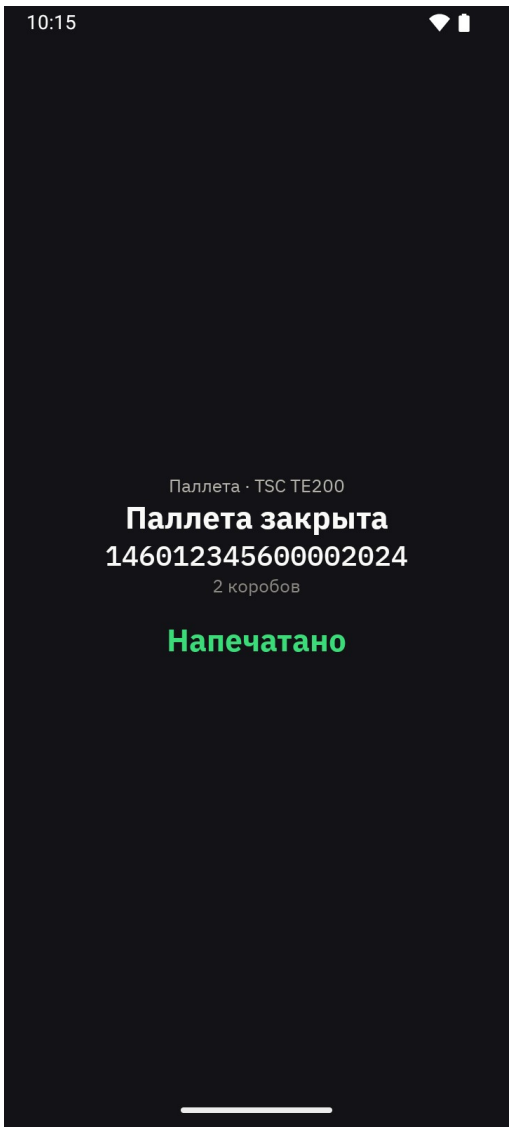
When the number of boxes reaches the pallet capacity, the handheld closes the pallet itself. To close earlier, press «Закрыть палет досрочно» (Close pallet early) and confirm with «Закрыть палет» (Close pallet).



Handheld: closing a pallet early

Step 5. Apply the pallet label

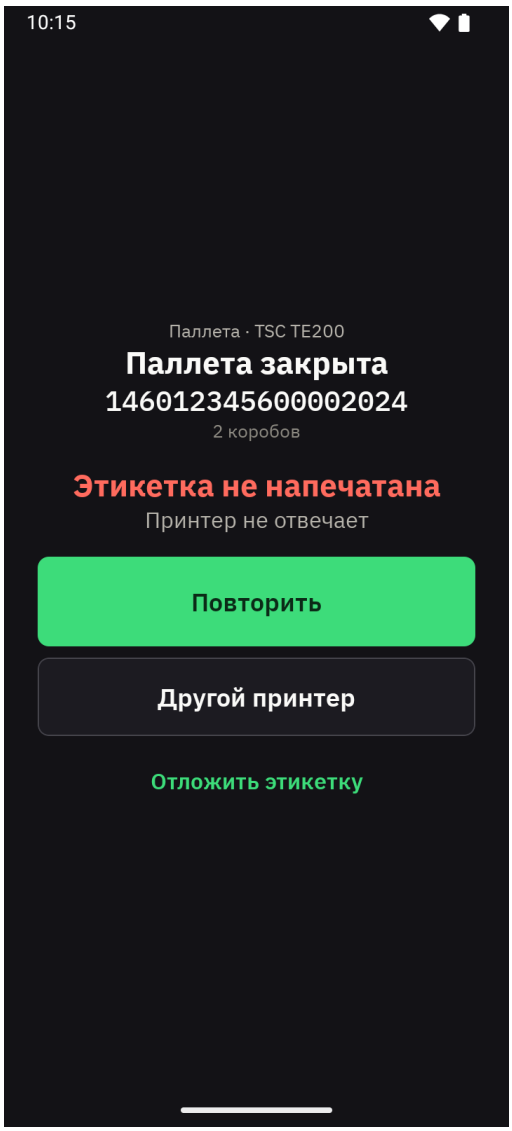
After closing, the handheld prints the pallet label on the printer assigned to «Палет» (Pallet): the «Палет закрыт» (Pallet closed) screen shows the pallet SSCC, the number of boxes and «Напечатано» (Printed). Check that a physical label came out, compare its SSCC with the screen and apply it to the pallet. The handheld then waits for the next pallet's first box.



Handheld: the pallet is closed and its label printed

Step 6. If the label did not print

On a print error the handheld says «Этикетка не напечатана» (Label not printed) and the reason. Fix it and press «Повторить» (Retry), or choose «Другой принтер» (Another printer). «Отложить этикетку» (Postpone label) leaves the label for later: the main screen gets a line about unprinted labels that opens a list with «Напечатать все» (Print all). If the handheld says «Результат печати неизвестен» (Print result unknown), look at the printer: if the label came out, press «Этикетка напечаталась» (The label printed), otherwise «Напечатать ещё раз» (Print again).



Handheld: the printer does not respond

Above the closure result, the screen names the printer recorded for this pallet. Earlier versions could show «Не назначен» (Unassigned) even when printing was configured; older pallet data may also lack the name. Check the «Палет» (Pallet) assignment in settings and the actual label at the printer.

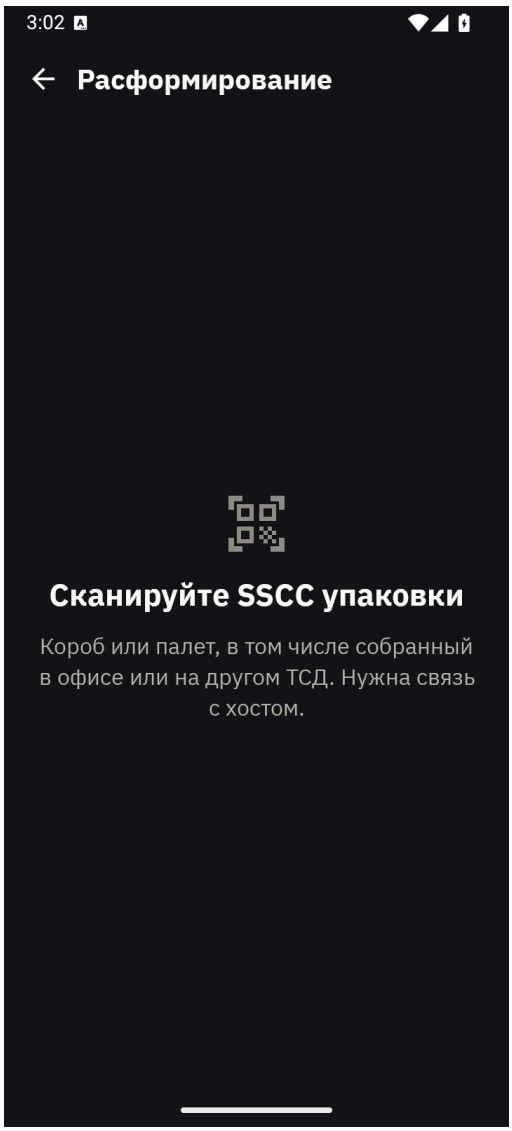
Important: If the host did not accept a box, the «Палеты» (Pallets) screen shows «Снимите с паллеты N короба» (Remove N boxes from the pallet) with the boxes and the reason. Remove those boxes from the pallet and press «Принято» (Understood); if the pallet is already closed and labelled, press «Перепечатать этикетку» (Reprint label) and replace the label.

7. Disassembling a pack

Open «Расформирование» (Disassembly) on the main screen. This mode requires a host connection. It accepts a closed box or pallet from the site, including one built on another device. If the pack is from a production shift, close that shift first.

Step 1. Scan the SSCC

On «Сканируйте SSCC упаковки» (Scan pack SSCC), scan the label. Check the product, SSCC and box or code count. Choose a reason from the host's list. Contact the office if the list is empty.



Handheld: scanning a pack SSCC

Step 2. Confirm and wait for the document

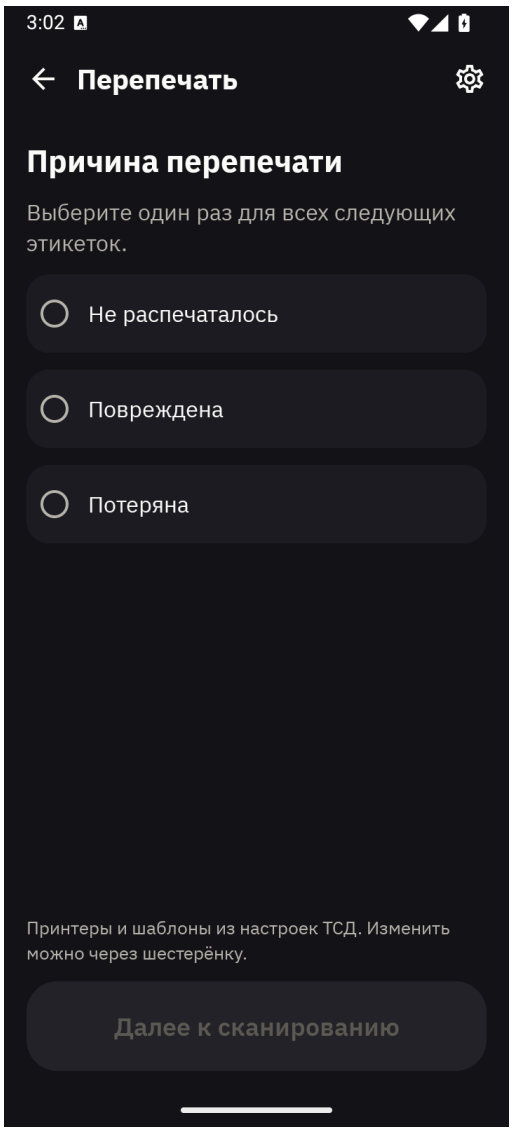
Tap «Расформировать» (Disassemble). When «Упаковка расформирована» (Pack disassembled) appears, remove the old label. Disassembling a box releases its codes. Disassembling a pallet releases its boxes with their own SSCCs. The document number appears on the handheld. The report and XML are available in the office. Refresh the registry on other handhelds before building pallets from the released boxes.

Important: If the handheld shows «Результат пока неизвестен» (Result still unknown), restore the connection and tap «Проверить результат» (Check result). The handheld checks the saved request. Confirm the result or check the pack's status in the office before starting another operation.

8. Reprinting a label

Step 1. Choose a reason

Open «Перепечать» (Reprint) on the main screen. Choose a reason for this session and tap «Далее к сканированию» (Continue to scanning). The gear on the reason screen opens printer and template settings.



Handheld: choosing a reprint reason

Step 2. Find the label by its code

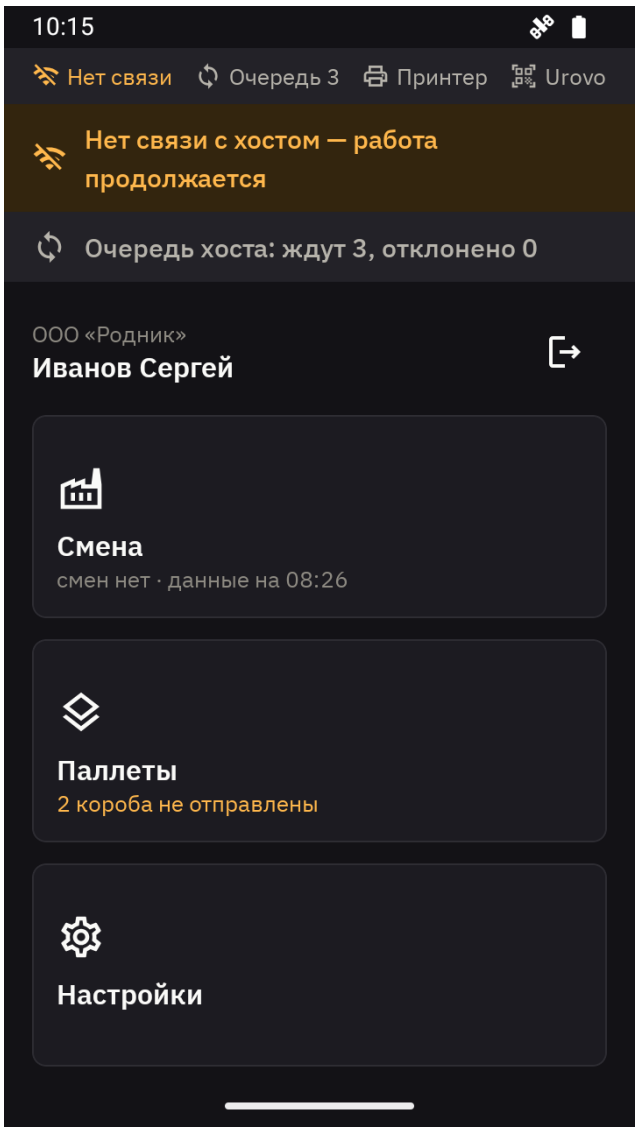
While connected to the host, scan a complete marking code or a box or pallet SSCC. To type the code, tap «Ввести код вручную» (Enter code manually), enter the complete code and tap «Найти и напечатать» (Find and print). Printing starts as soon as the code matches. Disassembled packs and released codes cannot be reprinted.

Important: If the print result is unknown and a label came out, tap «Проверить этикетку» (Verify label) and scan the printout. Verification does not print another label. If there is no printout, choose «Напечатать ещё раз» (Print again) and read the warning about a possible duplicate.

9. Working without a link to the host

Step 1. Keep working

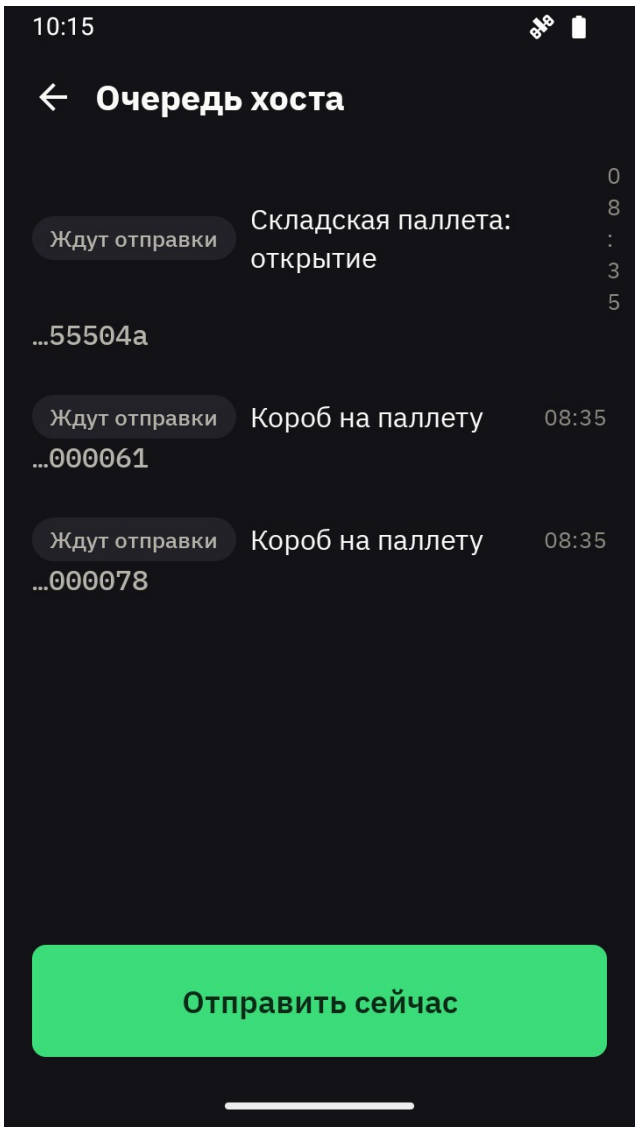
Without a link the handheld shows «Нет связи с хостом — работа продолжается» (No link to the host — work continues). Only boxes from the previously downloaded registry are available; closing pallets requires remaining SSCC numbers. If the numbers run out, reconnect to the host: the pallet remains open. Within these limits, pallet building and label printing go on: records collect on the handheld, and the «Очередь хоста» (Host queue) line shows how many are waiting. The «Палеты» (Pallets) row shows how many boxes are unsent.



The handheld without a link to the host

Step 2. Check the queue

Tap the status indicator and choose «Открыть очередь хоста» (Open host queue). This screen lists the records that «Ждут отправки» (Wait to be sent) and those «Отклонено хостом» (Rejected by the host). When the link returns, the handheld sends the queue itself; «Отправить сейчас» (Send now) sends it without waiting. If the host rejected a record, press «Передать в поддержку» (Pass to support) and tell the administrator.



Handheld: the queue of records for the host

Important: Do not uninstall the app or reset the handheld while the queue is not empty: unsent records are stored only on this handheld.

10. Common questions

What you see	What to do
The «Смена» (Shift) row says «смен нет» (no shifts)	Check that the handheld has a line assigned and that the line has planned or active shifts. The administrator chooses «Линия» (Line) in the handheld's window under «Устройства» (Devices) and saves; the handheld receives it when connected to the host (step 3.7). If the field is absent, first update the office PC application. The warehouse «Палеты» (Pallets) mode also works without an assigned line.
«Справочник ещё не загружен» (Reference data not loaded yet) in «Палеты» (Pallets)	The handheld has not yet received products and templates from the host. Go back to the main screen and open «Палеты» (Pallets) again while the handheld is online with the host.
«Сертификат хоста изменился — нажмите, чтобы привязать ТСД заново» (The host certificate changed — tap to pair the handheld again)	The host presents a different certificate. In the office, in «Устройства» (Devices), press «Привязать заново» (Pair again) in this handheld's row; on the handheld tap the message and enter the new code. Data and the queue on the handheld are kept.

What you see	What to do
«Часы устройства расходятся с хостом — проверьте время на ТСД» (The device clock differs from the host — check the time on the handheld)	Set the same time and time zone on the handheld as on the host computer.
«Хост восстанавливается из копии — работа продолжается» (The host is being restored from a backup — work continues)	The administrator is restoring the host from a backup. Keep working: the queue goes to the host after the restore.
Check the handheld version or updates	In Settings, open «Об устройстве» (About device). Checking for and downloading updates requires internet access. Let the queue finish sending before installing, then confirm installation in Android.

Common questions

If a question is not resolved, tell the site administrator: the office shows the handheld, its version and its link to the host in «Устройства» (Devices).